ZIYU WANG

ziyuwang@berkeley.edu | (510) 646-3805 | Berkeley, CA 94704

EDUCATION

XI'AN JIAOTONG UNIVERSITY **Xi'an, China**
Bachelor of Science in Physics Sept. 2020 - June 2024

- **Average Grade:** 90.02/100; **GPA:** 3.82/4.3
- **Courses:** Mechanics (100/100), Electromagnetism (90/100), General Thermal Physics (96/100), Optics (91/100), General Physics-Atomic Physics (99/100), Theoretical Mechanics (98/100), Electrodynamics (94/100), Mathematical Methods in Physics (95/100)
- **Honors:**
  - **Finalist Prize** of COMAP's Mathematical Contest in Modeling (MCM)® (Top 2% of all participating teams worldwide) (2022)
  - School First Prize in Xi'an Jiaotong University (2022)
  - Outstanding student in Xi'an Jiaotong University (2022)
  - 2022 Provincial-level College Student Innovation Training Project - Excellent (2022)
  - 33rd Tengfei Cup School-level Excellence Award (2022)
  - Provincial First Prize in National Mathematics Contest (2021)

UNIVERSITY OF CALIFORNIA, BERKELEY **Berkeley, CA**
Exchange Student Jan. 2023 – Aug. 2023

- **GPA:** 4.00/4.00
- **Courses:** Quantum Mechanics (A+), Special Relativity and General Relativity (A+), Solid State Physics (A+)

PATENTS

- Zhou, Z., **Wang, Z.**, Tang, Z., et al. (2023). A plant-wearable self-powering system based on droplet frictional electricity and its method. Shaanxi Province: Patent Application No. CN115967297A

RESEARCH EXPERIENCE

RESEARCH ON SUPERHYDROPHOBIC TRIBOELECTRIC NANOGENERATORS **Xi'an, China**
*Advisor: Prof. Zhifu Zhou, School of Energy and Power Engineering, **Xi'an Jiaotong University*** Sept. 2021 - May 2022

- Improved the performance of flexible triboelectric nanogenerators through surface treatment technology and Laser-Induced Graphene technology.
- Enhanced the energy storage process of flexible triboelectric nanogenerators using Micro-Supercapacitor.
- Designed a Rainwater Energy Harvester using flexible triboelectric nanogenerators and piezoelectric materials.
- Composed a literature review report summarizing the research progress on wearable triboelectric nanogenerators.
- Wrote a national patent as the first student-author.

BOGOLIUBOV QUASIPARTICLE INTERFERENCE IMAGING OF UNCONVENTIONAL SUPERCONDUCTORS **Notre Dame, IN**
*Advisor: Prof. Xiaolong Liu, Department of Physics & Astronomy, **Notre Dame University*** May 2022 – Aug. 2022

- Developed a MATLAB GUI to calculate the interference imaging of different materials.
- Compared calculated results with experimental observations.

RESEARCH ON MOIRÉ EXCITONS IN $MoSe_2/WS_2$ HETEROSTRUCTURES **Berkeley, CA**
*Advisor: Prof. Feng Wang, Department of Physics, **UC Berkeley*** Jan. 2023 – Aug. 2023

- Created devices of 2D materials and conducted optical measurements (absorption spectroscopy and photoluminescence measurements) on 2D devices and analyzed experimental data.
- Completed literature review of the MoSe2/WS2 system.
- Calculated the dipole-dipole interaction and exchange of excitons on moiré superlattices.

ACTIVITIES

RESEARCH ON THE PRIMARY AND SECONDARY SCHOOL ENROLLMENT SYSTEM IN XI'AN **Xi'an, China**
Social Practice Activity May 2021 – Aug. 2021

- Distributed questionnaires and conducted interviews with parents of primary and secondary school students, analyzed existing educational policies, and presented suggestions for improvement to the Education Bureau of Xi'an City

XI'AN JIAOTONG UNIVERSITY ABU-ROBOCON ROBOTICS TEAM **Xi'an, China**
A Member of Software Programming Group Aug. 2021 – Nov. 2021

- Wrote and uploaded programs to STM32 microcontrollers using Keil and J-Link software

SKILLS

- **Technical skills:** MATLAB, LaTeX, Mathematica, Python, C++, SQL
- **Languages:** Mandarin (native), English (fluent)